

Exercise 1 First Steps in Linux/Unix

Learning Objectives:

- A first orientation in the graphical interface of a Linux system
- Getting to know the standard shell of a Linux system
- Understanding and getting to know the Linux/Unix directory structure
- Creating, deleting, and moving directories and files
- Understanding and identifying the user and group concept in the system
- Displaying and changing file access permissions for users and groups

Legend:

 Reading material that must be read before the exercise

 Questions/tasks to be completed before the exercise

 Tasks to be completed during the exercise

1.1 Preparations for the Exercise

The preparations and questions should be completed before the exercise session. Exemption is Practice01. Here you have to take your time within the lesson. They are intended to assist you while completing the tasks during the exercise session.

 To prepare, study the at least one of the following materials

- <https://ubuntu.com/tutorials/command-line-for-beginners#1-overview>
- <https://kinsta.com/blog/linux-commands/>
- folien_bts_uebung_01_WS1314.pdf (in German)

 **Introductory Questions:**

Which command lists you the contents of a directory?

- ☐ pwd
- ☐ ls
- ☐ cdddfa

What does `pwd` stand for?

- ☐ Password Delete
- ☐ Present Working Directory
- ☐ Preserve Work Done

With which command can you display help pages in the shell under Linux/Unix?

Check all the commands with which you can fully display text files in the shell.

- ☐ `ls`
- ☐ `less`
- ☐ `mv`
- ☐ `more`
- ☐ `cat`
- ☐ `tac`
- ☐ `clear`
- ☐ `rm`

Permissions and Rights Read the introduction under `permissions-exercises.pdf` **only up to 1.) CHANGING FILE PERMISSIONS.**

What types of permissions are distinguished between user and group rights in Linux/Unix? What letters are used for the permissions? What bit is used for the permissions?

What is the command to make a file **readable and executable** only for the user? Provide this command both with arguments consisting of numbers and consisting of characters.



Important to Start:

First, we will jointly set up a Linux system on a Raspberry Pi. Alternatively, you can set up a virtual machine with a Linux distribution on your private non-Linux system. However, we cannot offer support for this.



1.2 Files and Directories

- Change to the root directory and display all the visible directories there.
- Create a directory `uebung01` in your home directory.
- Create a text file `wurzel.txt` in the `uebung01` directory in your home directory on the Linux system.
- Write all the directories displayed above into it.
- Note in the text file which directory you think serves which purpose, and explain this in bullet points. You are free to choose from different editors to edit the file.
- Display the file `wurzel.txt` in the shell.
- What is the size of the file `wurzel.txt`?
- What access permissions does the file `wurzel.txt` have? (perhaps already look at the next task)
- Remember to copy the file `wurzel.txt` to your server directory.
- Delete the file `wurzel.txt` and the directory `uebung01` but only after copying it to your server space.

1.3 File and Directory Permissions

- a) Create a file `working.txt` with a command in your home directory!
- b) Change the file permission with `chmod 444 working.txt`. What does this command do? What permissions does the file have?

- c) Now only the user should be able to write to the file. Execute the appropriate command in the shell.
- d) Remove permissions with `chmod u-r working.txt`
- e) Make the content visible with `cat working.txt`. What do you see?

- f) How could you restore the permissions now?
- g) Delete the file `working.txt` from your home directory at the end of the exercise.